Cutnell & Johnson
Newton's Third Law of Motion Part 1
Collisions
Physics
Sixth Edition
PASPort®

Student Information

§ Enter the appropriate information in the boxes below.

Date:	
Your Name:	
Group:	
Period:	

Introduction *Exploration Goals*

§ The purpose of this exploration is to determine the forces exerted on two objects in a collision. You will measure and compare the forces on each of the two bodies.

§ For this activity, use Force Sensors attached to carts to measure the force of interaction during a collision between the two carts.

Learning Outcomes

You will be able to:

• Compare the force exerted on one cart during a collision to the force exerted on the second cart during the collision.

• Use a graph of force versus time for each cart to determine the force exerted on each cart during a collision.

• Describe Newton's Third Law of Motion in conceptual terms.

• Demonstrate a quantitative understanding of Newon's Third Law by applying it to a problem related to collisions.

Hypothesis

§ How does the force exerted on one cart during a collision compare to the force exerted on the second cart in the collision?

§ If the carts have different masses, how will the force exerted by the heavier cart on the lighter cart compare to the force exerted by the lighter cart on the heavier cart?

Background

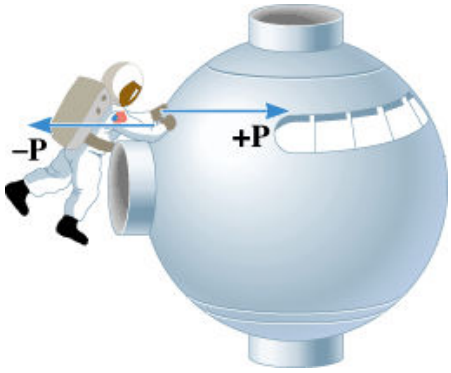
§ Whenever one body exerts a force on a second body, the second body exerts an oppositely directed force of equal magnitude on the first body.

$$F_{12} = -F_{21}$$

Newton's Third Law

§ The third law is often called the “action-reaction” law, for it is sometimes quoted as follows: “For every action (force) there is an equal, but opposite, reaction.”

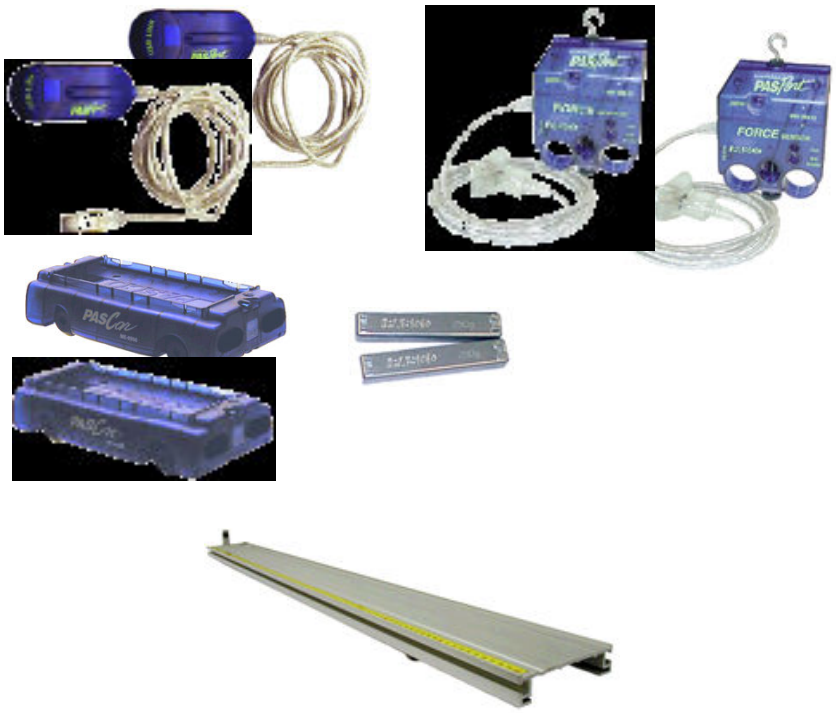
§ The diagram illustrates how the third law applies to an astronaut who is drifting just outside a spacecraft and who pushes on the spacecraft with a force P. According to the third law, the spacecraft pushes back on the astronaut with a force -P that is equal in magnitude but opposite in direction.



» For more information see Cutnell & Johnson, Physics, 6th ed., Volume One, Chapter 4, Section 4.5.

Materials

- 2 PASPORT USB Links
- 2 PASPORT Force Sensors
- 2 PAScars
- Mass Bars
- 1.2 m Dynamics Track



Setup *Equipment Setup*

Safety Reminder: Follow the directions for using the equipment.

§ Place the Dynamics Track on a horizontal surface. Level the Dynamics Track by placing the PAScar on the Dynamics Track. If the cart rolls one way or the other, use the Adjustable Foot at one end of the Dynamics Track to raise or lower that end until the Dynamics Track is level and the cart does not roll one way or the other.

§ Use the thumbscrew that comes with the Force Sensor to mount the sensor onto the accessory tray of each cart.

§ Remove the hooks from the Force Sensors.

§ Attach the cylindrical rubber bumper that comes with the Force Sensor to the front of each sensor.

Note: It is very important that the track is level to get the best results.

Process *Prepare to Record Data*

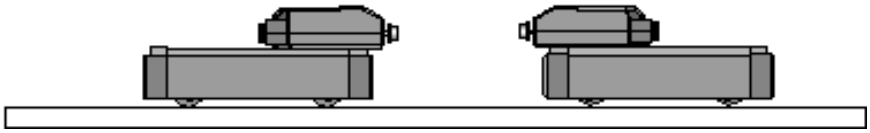
» The idea is to push the two carts together so they will collide. The Force Sensors will measure the forces exerted on each cart during the collision.

§ After you measure the forces during a collision for carts of equal mass, add 500 g of mass to one cart and then repeat the collision.

» **Note**: Do not push the carts together too fast or they may slip off the track during the collision.

» The graph of force versus time is set up so that the measurements from the two sensors will be plotted on opposite sides of the X-axis. (One sensor is "Force, push right" and the other sensor is "Force, push left".) The area under each curve corresponds to the force.

» It is important to **Zero** both of the Force Sensors prior to each data run. When the



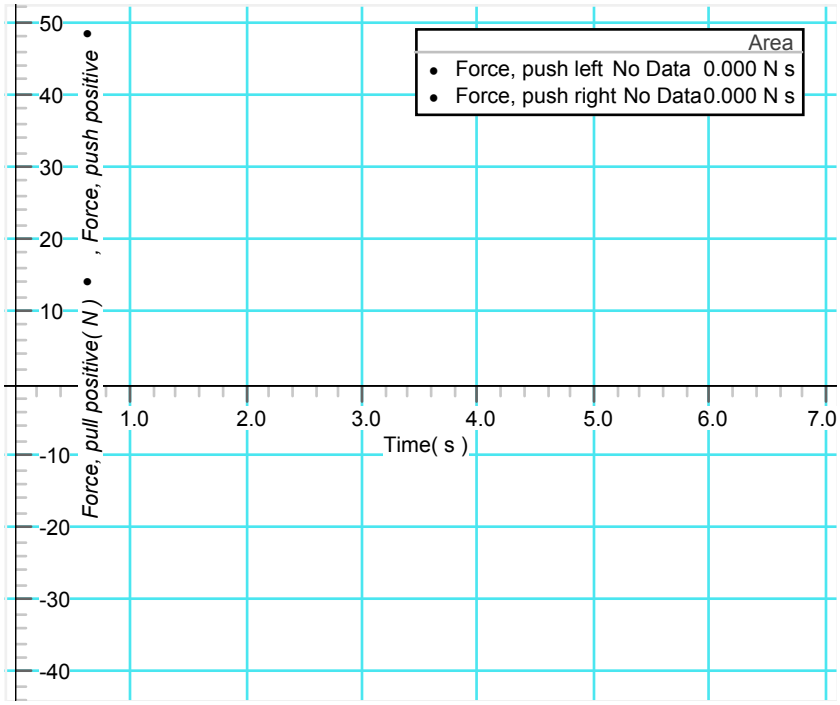
Process *Record Data*

§ **Zero** the sensors. Click **Start**. Push the carts together. Click **Stop** after the collision.

» The next page shows how to analyze your force data.

§ For Run #2, add 500 grams of mass to one of the carts.

§ **Zero** the sensors. Click **Start**. Repeat the data recording procedure again, but allow the lighter cart to remain stationary and push only the heavier cart. Click **Stop** after the collision.



Analyze

How to Find the Amount of Force

§ Click the **Scale to Fit** button if needed.

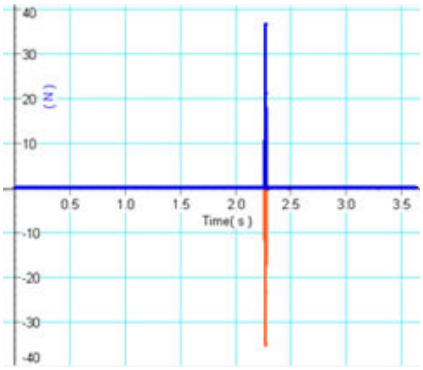
§ Use the **Zoom Select** tool to select a region around the part of Run #1 that shows the collision.

» The region you select will rescale to fit the graph.The Legend box shows the Area for each cart for Run #1.

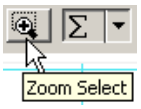
§ Record your area values on the next page.

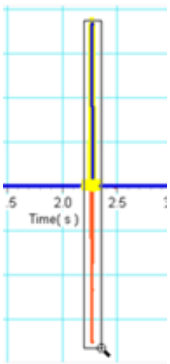
§ To analyze Run #2, click "Force, push right" in the Legend box. Use the **Data** menu to select Run #2. Repeat the procedure for "Force, push left".

§ Use **Zoom Select** as before.

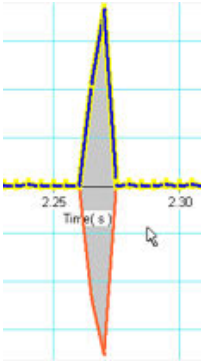


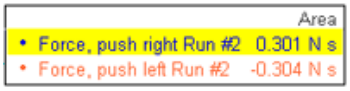
Sample data





Select a region of data





Click the Legend box to select the sensor, then use the **Data** menu to select Run #2.

Analyze

Find the Amount of Force

§ Use the procedure described previously to find the Area under the curve for each sensor for Run #1.

§ Repeat for Run #2.

Run #1

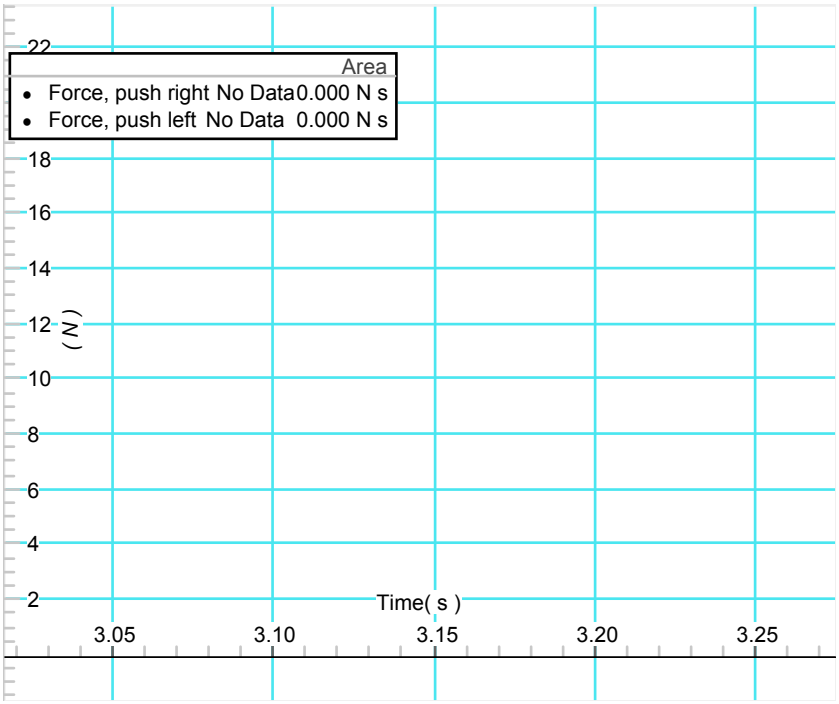
Force, push right: Area

Force, push left: Area

Run #2

Force, push right: Area

Force, push left: Area



Observations

§ What did you observe about the shape of the graphs of force versus time for each sensor?

§ What happened to the shape of the graphs of force from Run #2 when the heavier cart collided with the lighter cart?

Analyze *Comparison*

§ Which cart experiences more force in a collision when both carts are moving and have equal masses?

§ Which cart experiences more force in a collision when a heavier cart that is moving collides with a lighter cart that is not moving?

§ What effect does the mass of a cart have on the force it experiences?

Run #1

Force, push right: Area

Force, push left: Area

Run #2

Force, push right: Area

Force, push left: Area

Synthesize *Conclusions and Error Analysis*

§ How does the force exerted on one cart during a collision compare to the force exerted on the second cart in the collision?

§ If the carts have different masses, how will the force exerted by the heavier cart on the lighter cart compare to the force exerted by the lighter cart on the heavier cart?

§ How does your hypothesis compare to your results? Hypothesis:

Synthesize *Applications*

§ What are examples of events where Newton's Third Law apply?

Synthesize *Extension Problem*

§ In Cutnell & Johnson, Physics, 6th ed., Volume One, Chapter 4, Problem 14, pp. 116-7.

Airplane flight recorders must be able to survive catastrophic crashes. Therefore, they are typically encased in crash-resistant steel or titanium boxes that are subjected to rigorous testing. One of the tests is an impact shock test, in which the box must be able to survive being thrown at high speeds against a barrier. A 41-kg box is thrown at a speed of 220 m/s and is brought to a halt in a collision that lasts for a time of 6.5 ms (milliseconds). What is the magnitude of the average net force that acts on the box during the collision?

Notes